

NAME:	Date:
SECTION:	Professor:
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Configuring STP

Objectives

- View the default configuration of the spanning tree protocol.
- Select and enable the root bridge.
- Enable portfast on ports connected directly to host devices
- View how these configurations change the way in which traffic is forwarded.

Background/Scenario

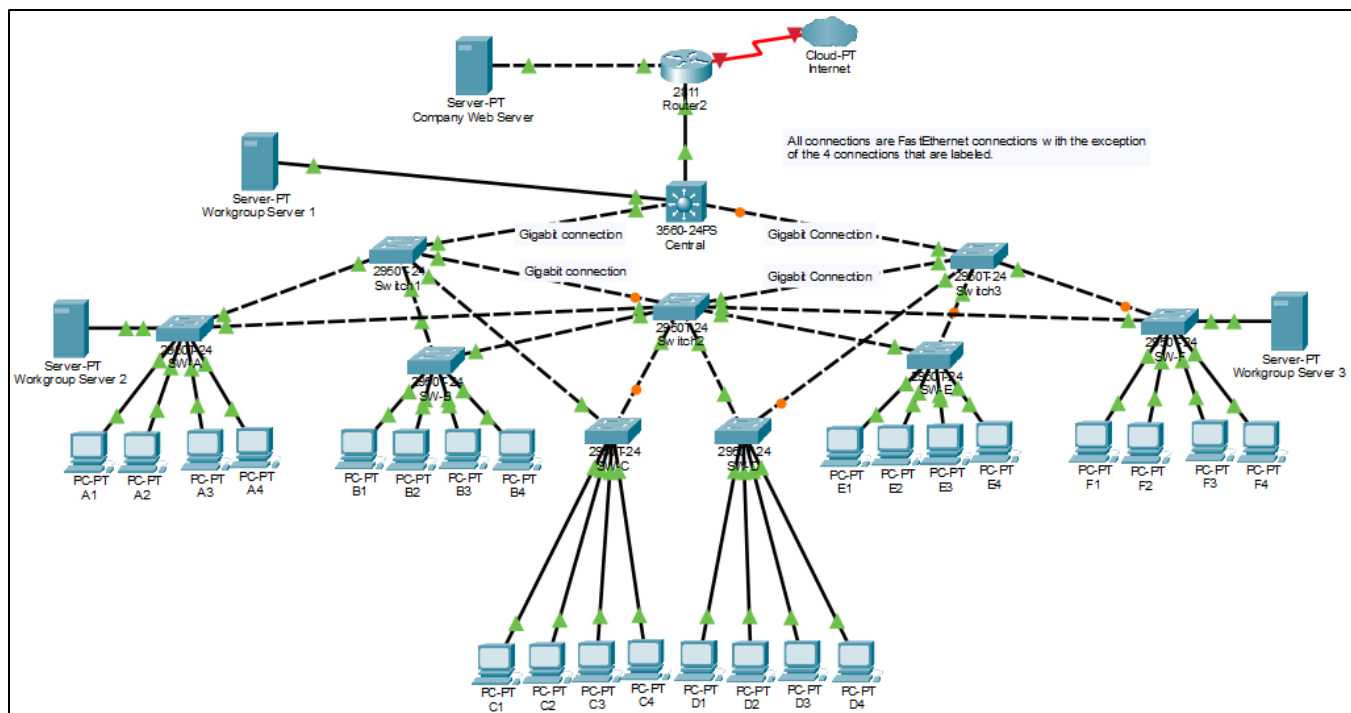
The spanning-tree protocol is useful in ensuring that loops do not exist in the network. However, it can introduce increased latency and unintended inefficiencies if left to auto negotiate the root bridge. In this activity, observe how the spanning-tree protocol affects path selection and how that path selection can change based on the configuration of the root bridge.

Task 1: View the Default Configuration.

Step 1. Wait for STP convergence to occur.

Spanning-tree must converge before connectivity occurs on a network. Wait for all link lights to turn either green or orange (green for forwarding, orange for blocking).

Observation: All ports take equally as long to converge, including those ports that are connected to host devices.



Step 2. Determine the root bridge.

- On the Central switches, enter privileged EXEC mode using the console password **cisco** and the secret password **class**.
- From privileged EXEC mode, issue the **show spanning-tree** command and observe the output.

In 3560-24PS Central Switch:

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0000.0C19.7455
             Cost        23
             Port        25 (GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     00D0.D31C.634C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface       Role Sts Cost      Prio.Nbr Type
-----
Gi0/2           Altn BLK 4         128.26   P2p
Gi0/1           Root FWD 4         128.25   P2p
Fa0/1           Desg FWD 19        128.1    P2p
Fa0/13          Desg FWD 19        128.13   P2p

Switch#
```

Observation: Central is NOT the root bridge. Notice that Fa0/13 is placed in designated forwarding, Gi0/2 is placed in alternate blocking, and Gi0/1 is specified as Root Fwd. Gi0/1 is connected to Switch1.

- c. On Switch1, from the privileged EXEC mode, issue the **show spanning-tree** command.

In 2950T-24 Switch1:

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0000.0C19.7455
             Cost        19
             Port        22 (FastEthernet0/22)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0009.7C61.9058
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/20                   Desg FWD 19        128.20  P2p
Fa0/24                   Desg FWD 19        128.24  P2p
Fa0/22                   Root FWD 19        128.22  P2p
Gi0/1                    Desg FWD 4         128.25  P2p
Gi0/2                    Desg FWD 4         128.26  P2p

Switch#
```

Observation: Switch1 is NOT the root bridge. Notice that all ports are forwarding, and fa0/22 is specified as Root Fwd. Fa0/22 is connected to SW-B.

- d. On SW-B, from the privileged EXEC mode, issue the **show spanning-tree** command.

In 2950T-24 SW-B:

```
switch5#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0000.0C19.7455
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0000.0C19.7455
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

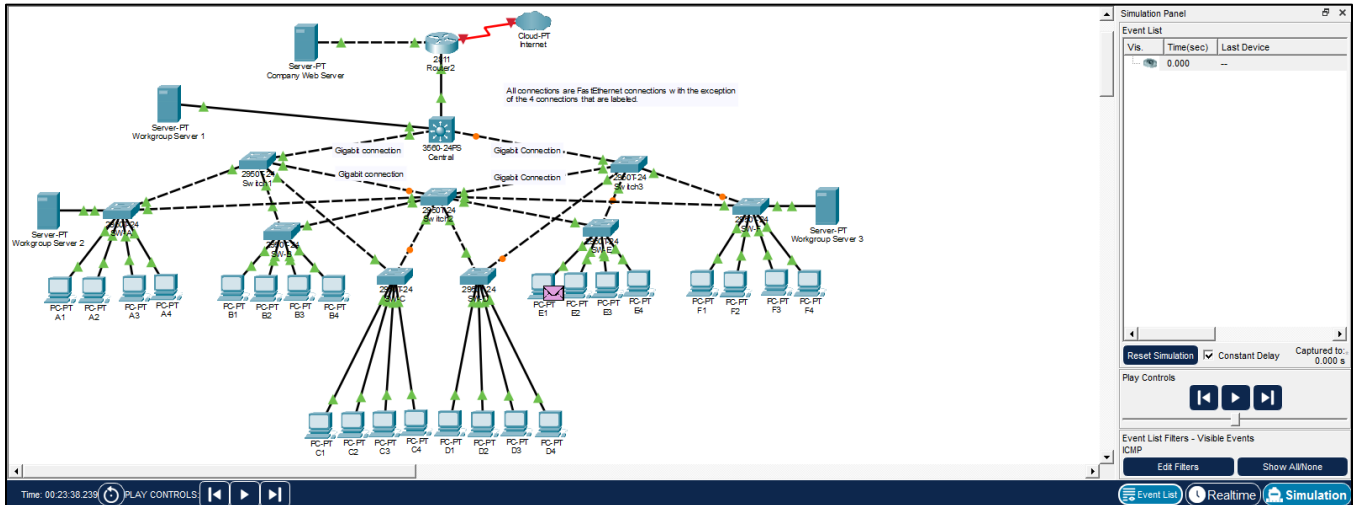
Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/22                   Desg FWD 19        128.22  P2p
Fa0/24                   Desg FWD 19        128.24  P2p
Fa0/4                    Desg FWD 19        128.4   P2p
Fa0/1                    Desg FWD 19        128.1   P2p
Fa0/2                    Desg FWD 19        128.2   P2p
Fa0/3                    Desg FWD 19        128.3   P2p

switch5#
```

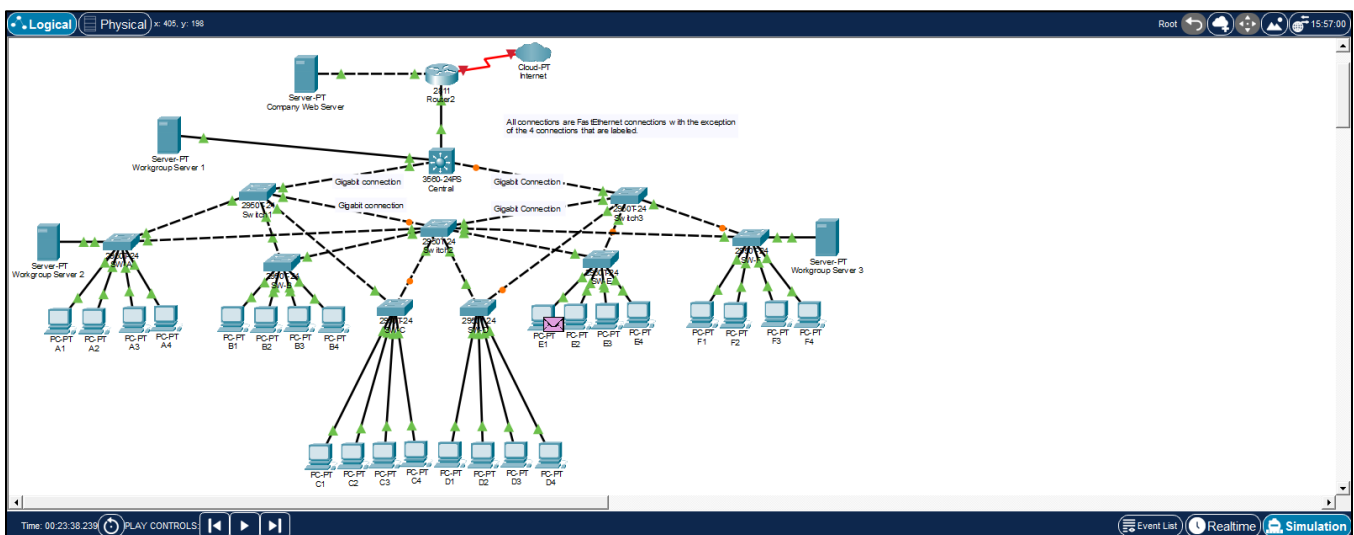
Observation: SW-B is the root bridge. This is evident by the line “This bridge is the root”.

Step 3. View how traffic is forwarded based on SW-B as root.

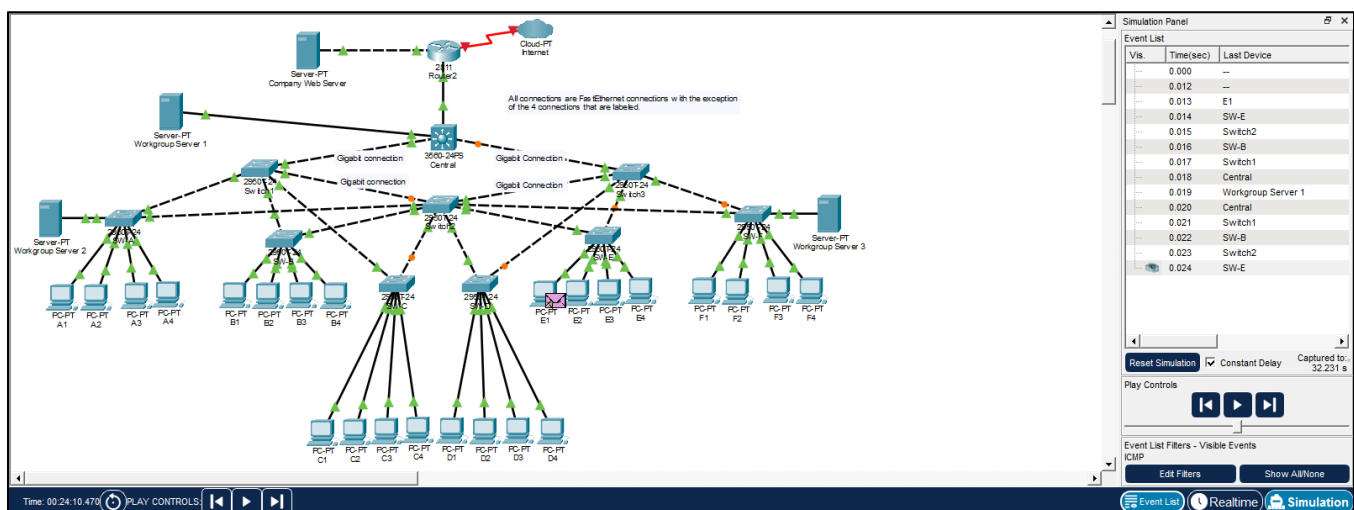
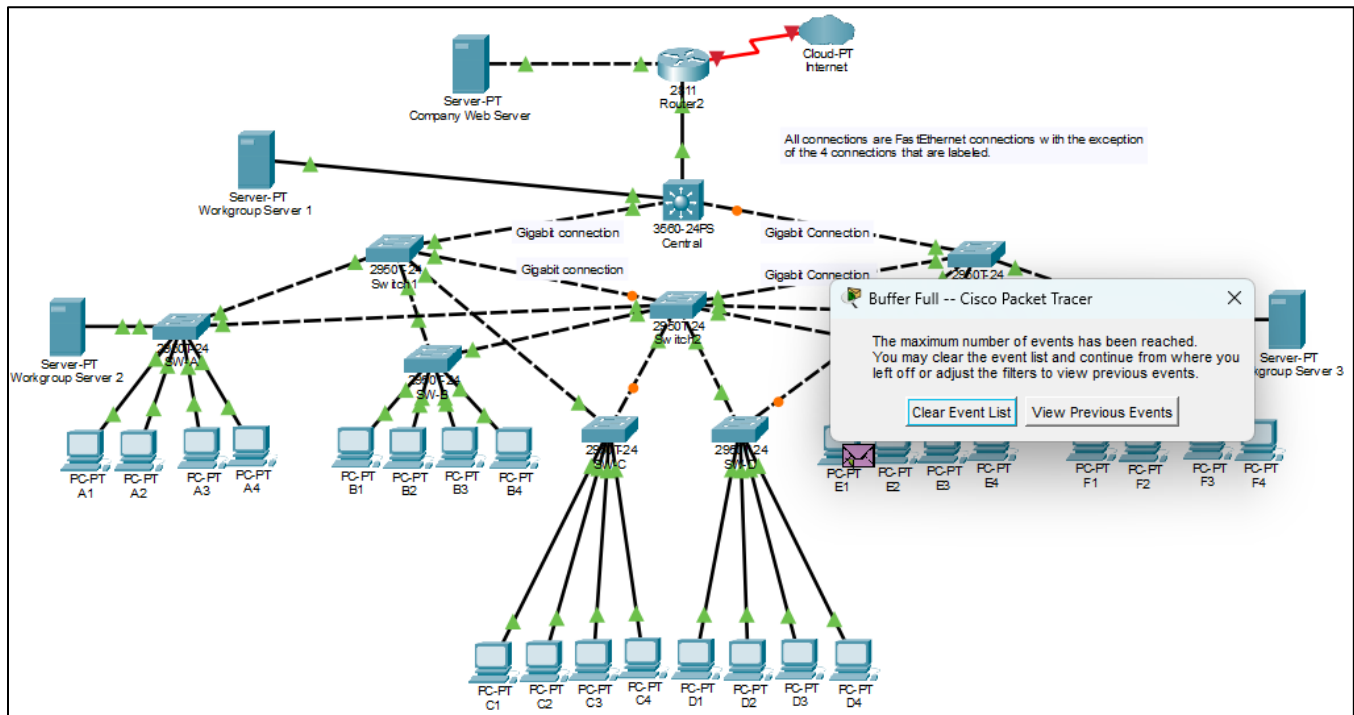
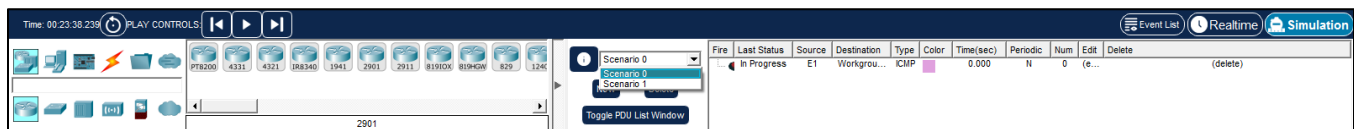
- e. Click on the simulation button. The simulation button can be found on the lower right-hand corner of the PT window, behind the Realtime clock.



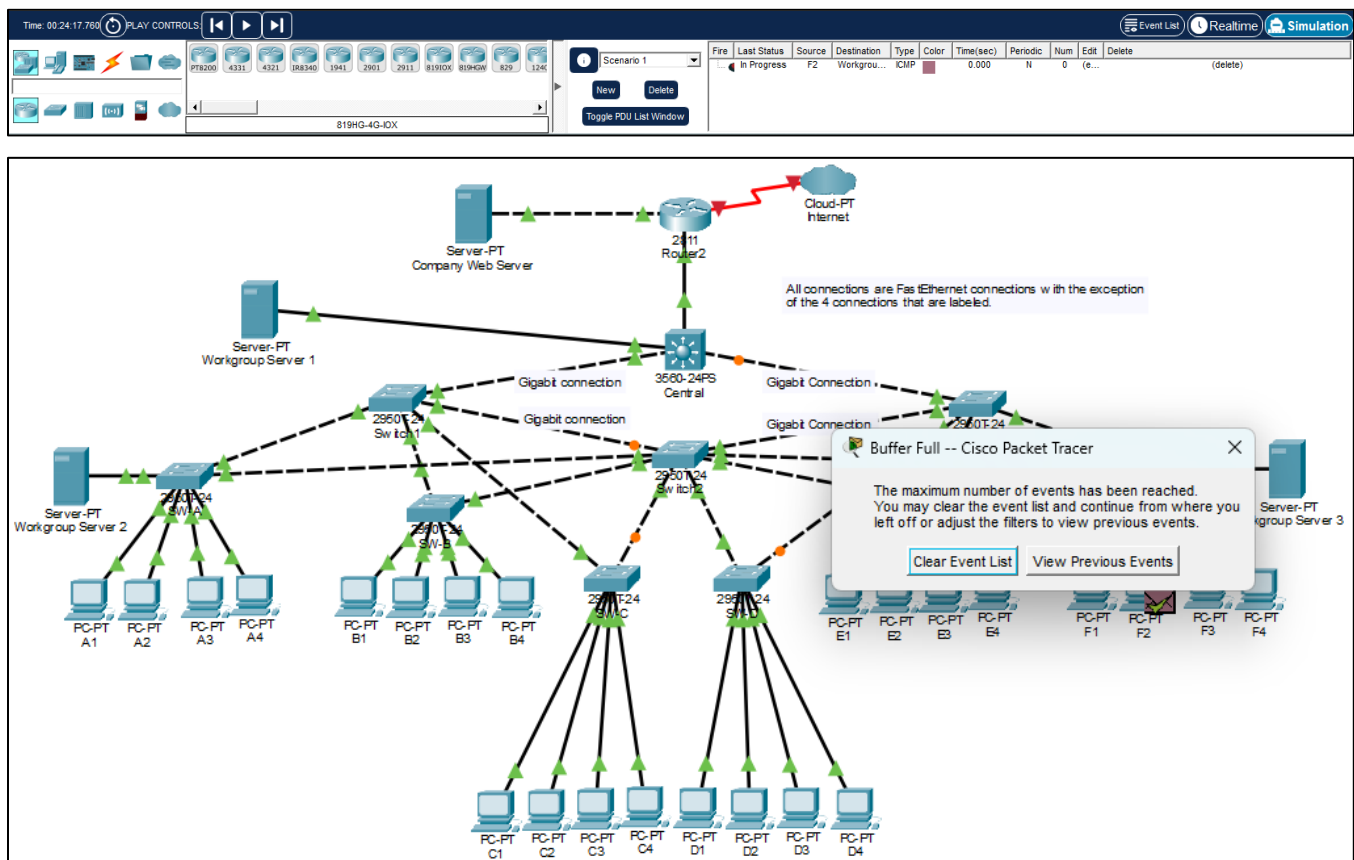
- f. Close the Event List window by selecting the x in the upper right-hand corner, just below the Viewport button. This will allow you to see the entire topology.

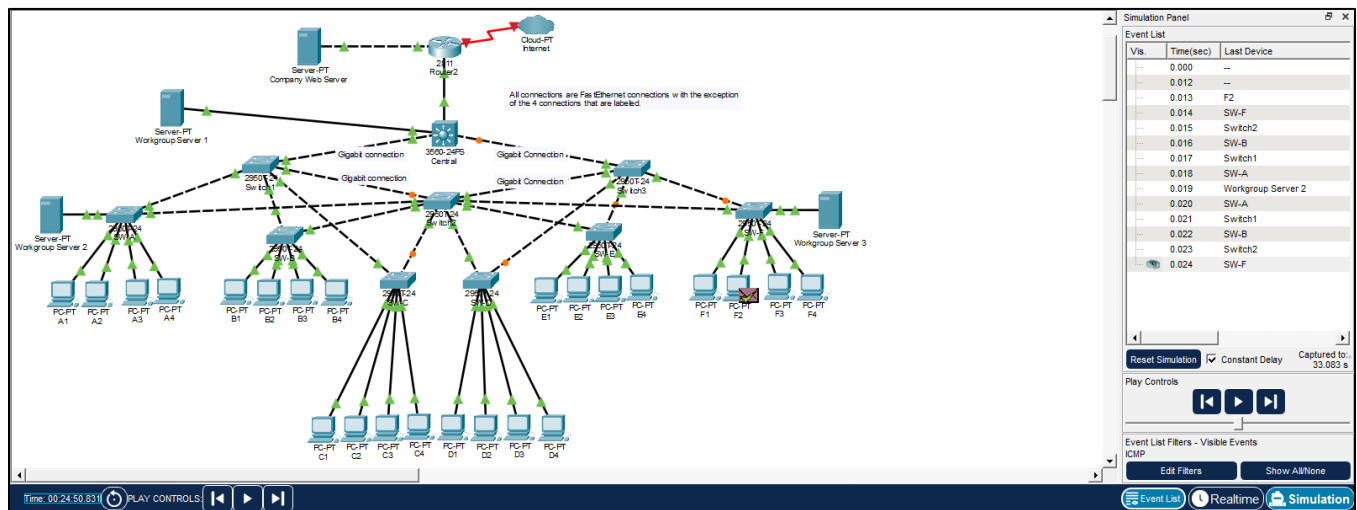


- g. Notice the Auto Capture / Play button and the Capture / Forward button on the bottom screen of the PT window. Below these buttons is a drop down box with Scenario 0 and Scenario 1. Be sure that Scenario 0 is selected. Scenario 0 is a ping originating from PC-PT E1 and destined to Server-PT Workgroup Server 1. Click the Auto Capture / Play button to view the path that the packet will take to reach that server. When the Buffer Full message appears, click on *view previous events*.



- h. Is the path that the packet took to reach that destination the most efficient path? Why did the packet take that path?
- **Is the path the most efficient?** No, it is not the most efficient path.
 - **Why did the packet take that path?** The packet is trying to reach Workgroup Server 1, which is connected directly to the main Central switch. The most efficient route would be to use the high-speed Gigabit links straight to Central. However, because SW-B is currently the root bridge, the Spanning Tree Protocol (STP) forces the packet away from Central, making it detour down to SW-B at the bottom edge of the network before it is allowed to travel back up to Server 1.
- i. Next, in the drop down box, select Scenario 1. Scenario 1 is a ping originating from PC-PT E2 and destined for Server-PT Workgroup Server 2. Click the Auto Capture / Play button to view the path that the packet will take to reach that server. When the Buffer Full message appears, click on *view previous events*.





- j. Is the path that the packet took to reach that destination the most efficient path? Why did the packet take that path?
- **Is the path the most efficient?** No, it is not the most efficient path.
 - **Why did the packet take that path?** The packet is trying to reach Workgroup Server 2, which is located on the far left of the network connected to SW-A. As your Event List screenshot perfectly proves, instead of taking a direct route across the middle switches (like Switch2 to SW-A), the packet is dragged all the way down to SW-B, pushed up to Switch1, and then finally sent down to SW-A. It takes this severely indirect path because STP blocks the faster, direct routes to prevent loops, forcing all traffic to pass through the root bridge (SW-B) first.

Step 4. Manually configure the root bridge.

- Click on the real time button. The real time button can be found on the lower right-hand corner of the PT window, behind the simulation timed clock.
- Select Central switch.
- From the EXEC privileged mode, enable the central switch as the root bridge by typing the command **spanning-tree vlan 1 root primary**.
- Save the configuration with the command **copy run start**.

In 3560-24PS Central Switch:

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spanning-tree vlan 1 root primary
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```


- e. Select Switch 2.
- f. From the EXEC privileged mode, enable the Switch 2 as the secondary root bridge by typing the command **spanning-tree vlan 1 root secondary**.
- g. Save the configuration with the command **copy run start**.

In 2950T-24 Switch2:

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#spanning-tree vlan 1 root secondary
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

Step 5. Enable Portfast.

To speed up convergence, it is also good practice to enable portfast on ports that are known to connect to servers and workstations. When an interface is configured with portfast, the STP for that port assumes that the port is not part of a loop and immediately moves to the forwarding state and does not go through the blocking, listening, or learning states. For this exercise, enable portfast on ports that are connected to servers.

- a. Click on SW-A
- b. Using the interface range command, enable portfast on ports Fa0/1 – Fa0/4. Additionally enable portfast on the port connected to Workgroup Server 2 (Fa0/13).

Example:

```
Switch(config)# interface range fa0/1 – fa0/4
Switch(config-if-range)# spanning-tree portfast
Switch(config-if-range)# interface fa0/13
Switch(config-if)# spanning-tree portfast
```

- c. Save the configuration with the **copy run start** command.

In 2950T-24 SW-A:

```
Switch4>ena
Switch4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch4(config)#interface range fa0/1 - fa0/4
Switch4(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
Switch4(config-if-range)#interface fa0/13
Switch4(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
Switch4(config-if)#exit
```

```
Switch4(config)#exit
Switch4#
%SYS-5-CONFIG_I: Configured from console by console

Switch4#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch4#
```

- d. Repeat steps a and b on switches SW-B, SW-C, SW-D, SW-E, and SW-F. In addition, on switch Central, enable portfast on port fa0/13.

In 2950T-24 SW-B:

```
switch5>ena
switch5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
switch5(config)#interface range fa0/1 - fa0/4
switch5(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
switch5(config-if-range)#interface fa0/13
switch5(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
switch5(config-if)#exit

switch5(config)#exit
switch5#
%SYS-5-CONFIG_I: Configured from console by console

switch5#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
switch5#
```

In 2950T-24 SW-C:

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1 - fa0/4
Switch(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if-range)#interface fa0/13
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#exit
```

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

In 2950T-24 SW-D:

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1 - fa0/4
Switch(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if-range)#interface fa0/13
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#exit
```

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

In 2950T-24 SW-E:

```
Switch4>ena
Switch4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch4(config)#interface range fa0/1 - fa0/4
Switch4(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
Switch4(config-if-range)#interface fa0/13
Switch4(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
```

```
Switch4(config-if)#exit
Switch4(config)#exit
Switch4#
%SYS-5-CONFIG_I: Configured from console by console

Switch4#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch4#|
```


In 2950T-24 SW-F:

```
switch5>ena
switch5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
switch5(config)#interface range fa0/1 - fa0/4
switch5(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
switch5(config-if-range)#interface fa0/13
switch5(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
switch5(config-if)#exit
```

```
switch5(config)#exit
switch5#
%SYS-5-CONFIG_I: Configured from console by console

switch5#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
switch5#
```

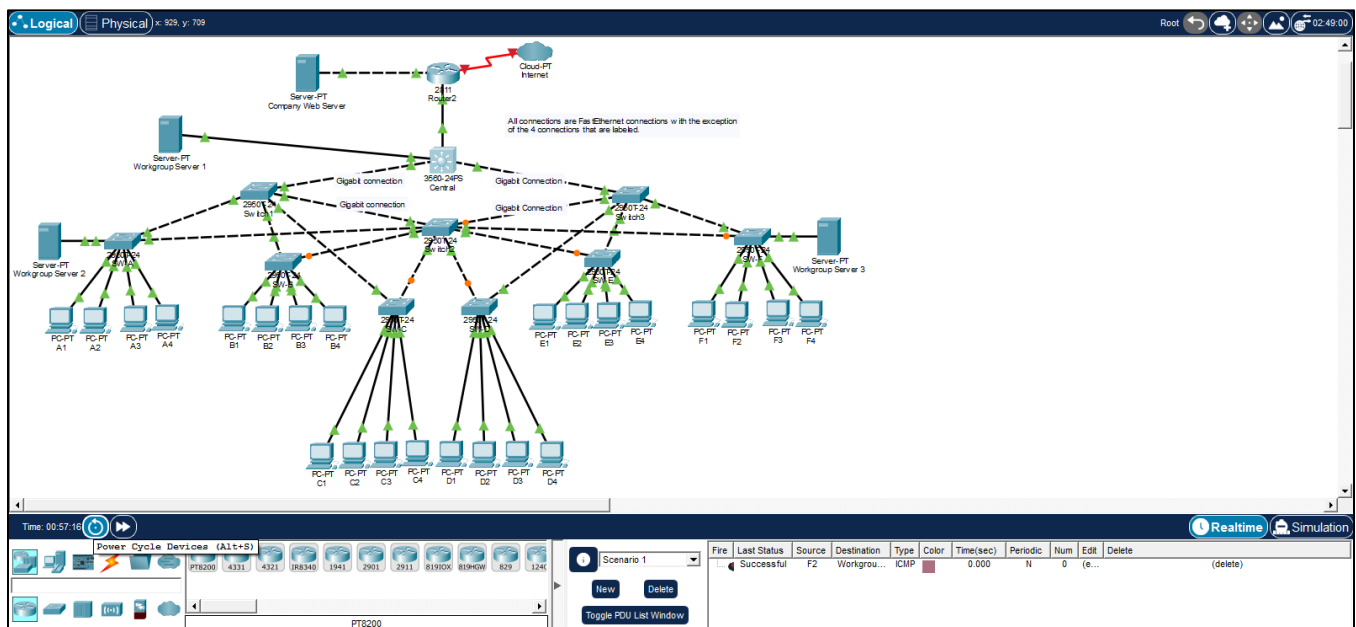
In 3560-24PS Central Switch:

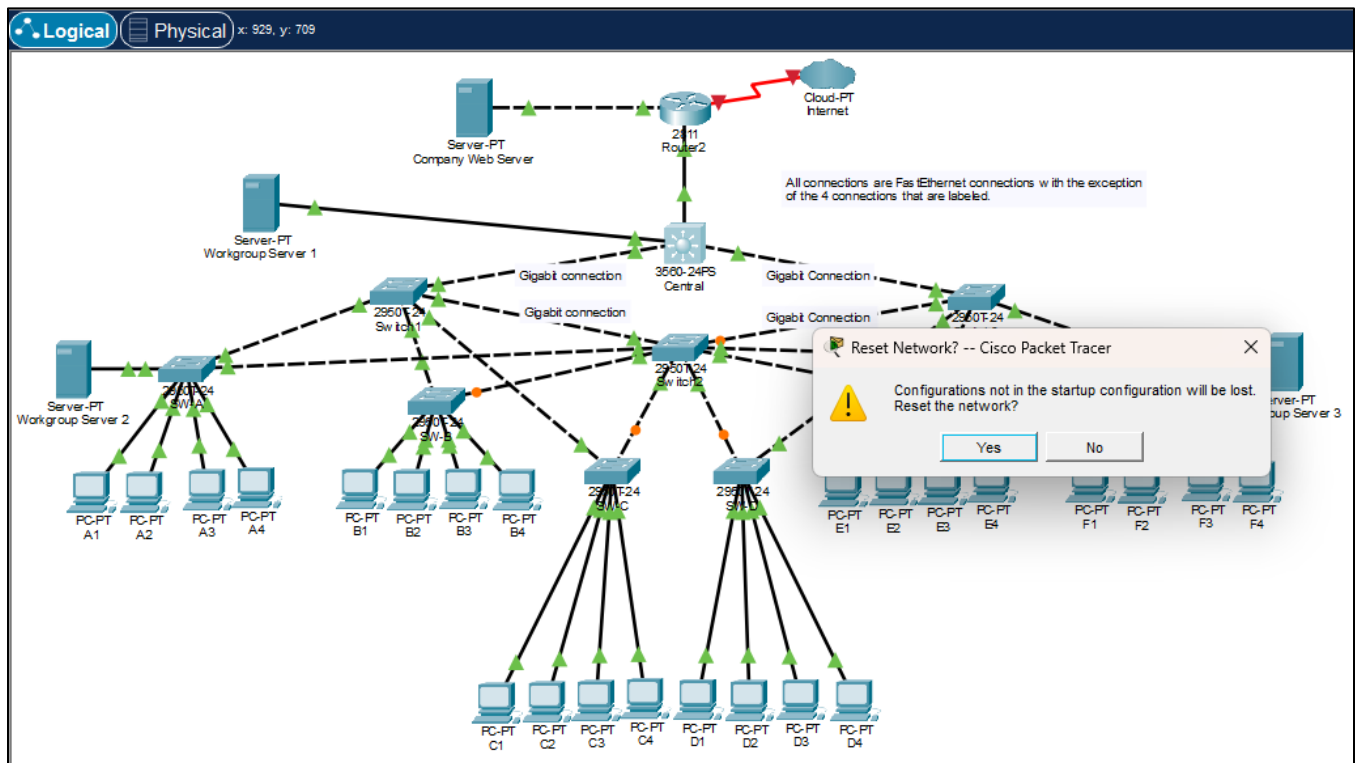
```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/13
Switch(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/13 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
```

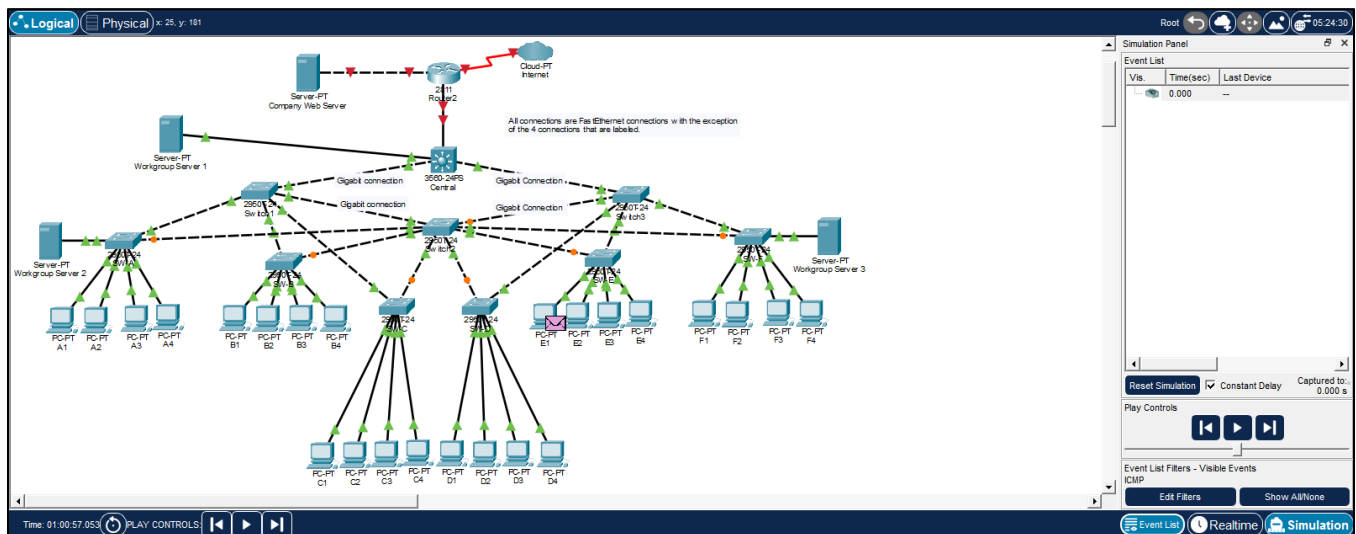
- e. Once you have completed the configuration, click the button “power cycle devices”. This will reboot all devices and allow you to watch the spanning-tree convergence. *Be sure that you saved all configurations before rebooting devices.*



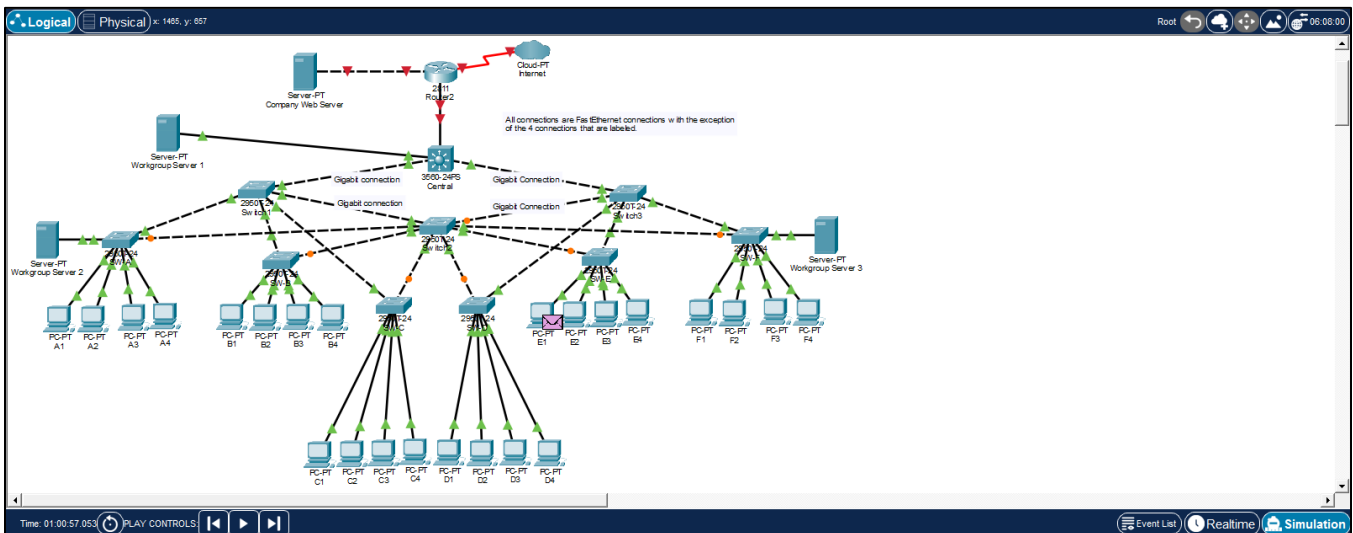


Step 6. View how traffic is forwarded based on Central as the root.

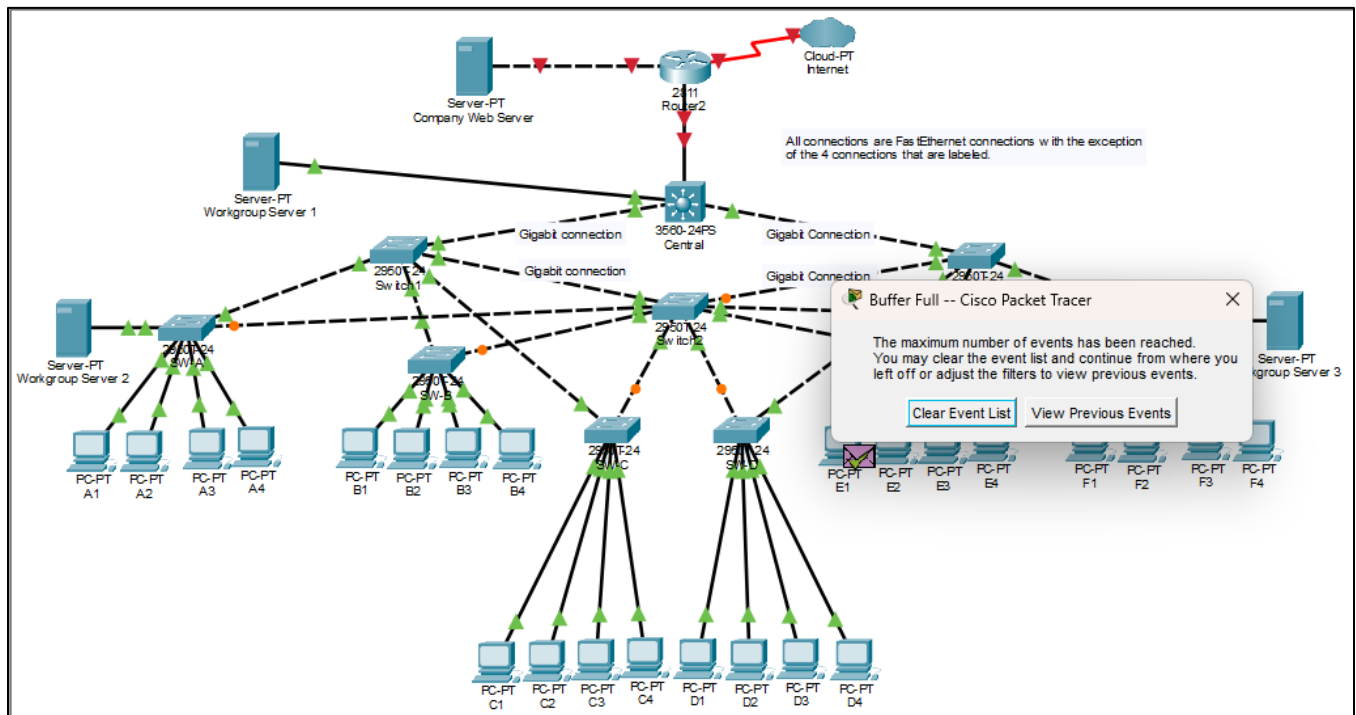
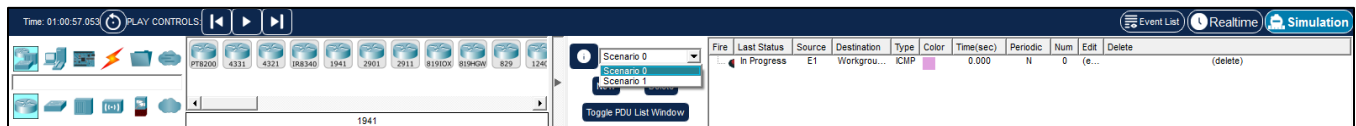
- Click on the simulation button.

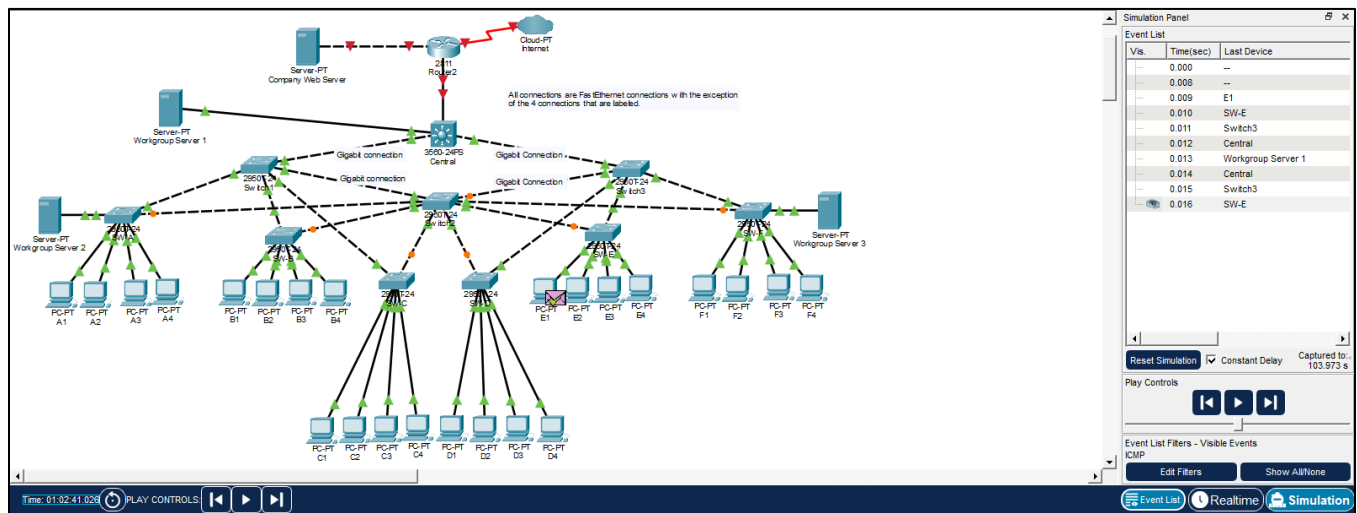


- b. Close the Event List window by selecting the x in the upper right-hand corner, just below the Viewport button. This will allow you to see the entire topology.



- c. In the drop down box, select Scenario 0. Scenario 0 is a ping originating from PC-PT E1 and destined to Server-PT Workgroup Server 1. Click the Auto Capture / Play button to view the path that the packet will take to reach that server. When the Buffer Full message appears, click on *view previous events*.

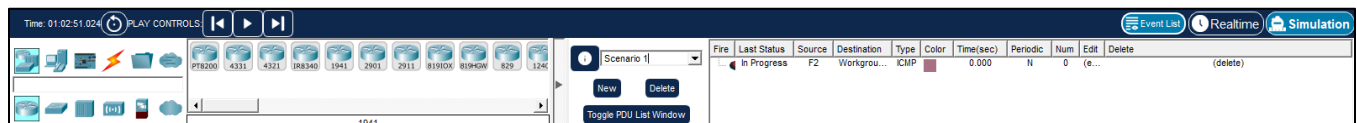


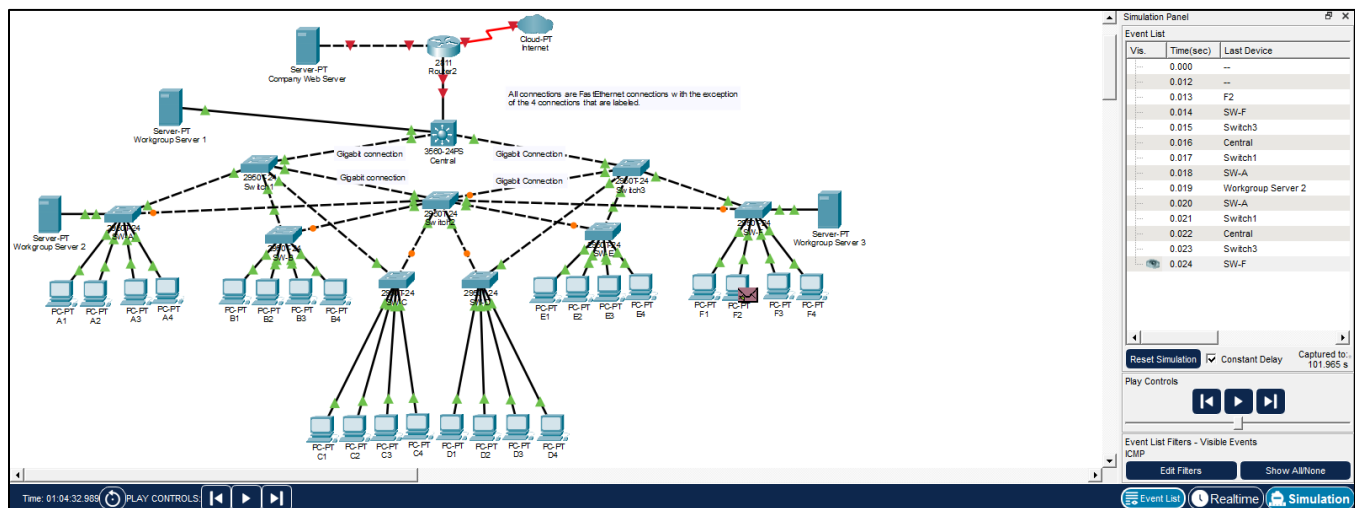
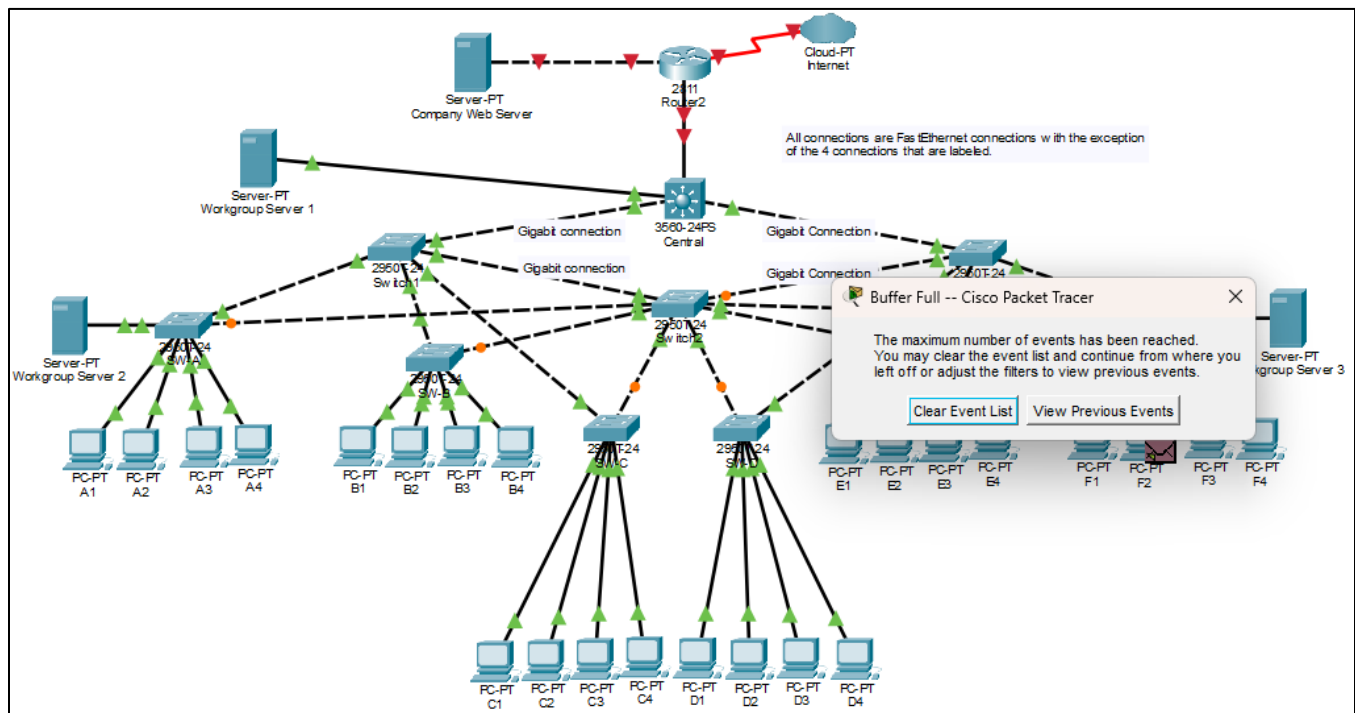


d. How has the path changed? Is the path more efficient? Is there more efficient paths available?

- **How has the path changed?** The packet now travels directly from SW-E through Switch3 to the Central switch, cutting out the previous detour through SW-B and Switch1.
- **Is the path more efficient?** Yes, it is much more efficient because the packet now takes the fewest number of hops possible and utilizes the high-speed Gigabit backbone links to reach the server.
- **Is there more efficient paths available?** No, this is currently the most direct physical and logical route available in the network topology for this scenario.

e. Next, in the drop down box, select Scenario 1. Scenario 1 is a ping originating from PC-PT E2 and destined for Server-PT Workgroup Server 2. Click the Auto Capture / Play button to view the path that the packet will take to reach that server. When the Buffer Full message appears, click on *view previous events*.





f. How has the path changed? Is the path more efficient? Is there more efficient paths available?

- **How has the path changed?** The packet now flows through the logical core of the network, traveling from SW-F to Switch3, then directly through the Central switch and Switch1 to reach its destination at SW-A.
- **Is the path more efficient?** Yes, it is much more efficient because the packet now utilizes the high-speed Gigabit backbone links and takes a more direct route through the center of the topology instead of detouring toward the old root bridge.
- **Is there more efficient paths available?** No, this is currently the most direct and efficient physical route available for these two devices now that the Central switch has been correctly configured as the root bridge.

Step 7. Reflection.

- a. How does specifying the root bridge affect efficiencies in the network?
 - Specifying the root bridge allows a network administrator to dictate the exact flow of data toward the core of the topology. By choosing a powerful central switch, you prevent the protocol from automatically electing a slower device that might create awkward detours. This control ensures that high speed backbone links are prioritized for most of the traffic moving between departments. It significantly reduces the number of hops a packet must take to reach its destination. Ultimately, this leads to a much more predictable and stable environment with lower overall latency.
- b. What are some factors that may be important to consider when selecting the root bridge?
 - We should primarily look at where the switch sits physically within your overall network hierarchy. A central location is ideal because it minimizes the logical distance to all other connected switches. Hardware performance is another critical factor since the root bridge must handle the heaviest management load. Link speeds also matter because you want your root bridge to utilize the fastest Gigabit connections available. Finally, consider which switch is closest to your most important resources like workgroup servers or internet gateways.
- c. In this topology there is only one VLAN and 1 root bridge. There is a separate instance of spanning-tree for each VLAN. How can specifying multiple VLANs and different root bridges for each VLAN improve efficiency?
 - This approach effectively enables load balancing by allowing different physical paths to be active at the same time. In a standard setup, many redundant links sit completely idle in a blocking state to prevent loops. By assigning different root bridges for each VLAN, you can make those idle links carry productive traffic. This distribution prevents any single backbone connection from becoming a bottleneck during peak usage hours. It maximizes the return on your hardware investment by utilizing every bit of available bandwidth.
- d. How does enabling portfast increase efficiency in the network?
 - Portfast increases efficiency by letting end user devices skip the long Spanning Tree convergence process. Normally, a switch port waits about thirty seconds to verify that no loops exist on that specific segment. Since we know a PC or server cannot create a loop, Portfast transitions the port to forwarding immediately. This prevents issues like DHCP timeouts where a computer gives up on getting an IP address before the port turns green. It creates a much smoother experience for users who need instant connectivity the moment they plug in.